

Bayesian machine learning for measurement noise estimation of focus variation microscopy instrument

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Abstract

ISO 25178-600 is an international standard explaining metrological characteristics that are applied to calibrate the optical areal surface topography measuring instruments. The determination of measurement noise is typically one of the metrological characteristics to be estimated for calibrating areal surface topography measuring instruments. However, the metrological characteristic estimation, that is, specific to the vertical heights of the optical instrument stage, requires calibration time. Commonly, the estimation of measurement noise is only applied for a specific range from the total vertical range of an instrument and measurement parameters. The noise can significantly affect and increase the measurement uncertainty of measurements. Reducing measurement noise is crucial to reduce the overall measurement uncertainty and then improve the measurement reliability. Calibrating the measuring instruments can help quantify this noise contribution to the uncertainty and hence can develop good measurement practices on how to minimize or reduce the measurement noise. Machine learning (ML) can estimate measurement noise for uncalibrated vertical axis heights by using known measurement noise from a specific vertical range and specific measurement parameters. This process of machine learning estimation can significantly reduce the overall time of performing the measurement-noise calibration for all vertical axis ranges of optical measuring instruments. This paper applies Bayesian machine learning to predict the measurement noise of a focus variation microscopy (FVM) on overall height locations based on measurement noise estimations from a few specific height locations of the instrument. This method can reduce the overall measurement operational cost. Different measurement parameters are used to quantify the measurement noise and the prediction test with the proposed machine learning method. The parameters include vertical and lateral resolutions, image contrast, exposure time, objective lens, illumination type and measurement position of the vertical motion stage axis. The measurement noise estimation is performed by measuring a standard flat-surface artefact and applying a subtraction method. The Bayesian machine learning estimation is optimized by finding the optimal hyper-parameter of the Bayesian model. Results show that Bayesian machine learning can predict unknown measurement noise at different vertical heights and with different measurement parameters at the accuracy up to 94%. In addition, the uncertainty of prediction is provided for reliable comparison with known data.

Keywords

machine learning, measurement noise, metrological characteristics, focus variation

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Introduction

The interest in using machine learning (ML) in advanced manufacturing has received significant attention over the last decade. The manufacturing processes of different sectors such as aerospace and automotive, have transferred four revolutions to the manufacturing industry of which data analysis and machine learning are a central component.¹ The transformation has led to making manufacturing facilities smart by connecting all processes, including measurement, with various manufacturing methods.^{1,2} One of the transitional operations is facilitated by applying digitization in

every industrial process including the works related to manufacturing metrology.³ The application of ML in manufacturing metrology is wide and the analysis of

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measuring instruments is part of these applications. The ML can be significantly leveraged to improve the efficiency and effectiveness of the measuring instrument's performance and reduce the overall cost of manufacturing metrology. Some recent examples of applying ML methods in manufacturing metrology can be found in Özel et al.⁴ and Liu et al.⁵ In these papers, ML methods are used in laser powder bed fusion (PBF) for surface texture parameter prediction and in an optical instrument to estimate the surface quality from scattered light data, that is, reflected from a measured surface.

As of today, optical (non-contact) measuring instruments are being used more and more in both industry as well as research laboratories as compared to the last decade. The main reasons for the high rate of adoption of optical instruments include their short measuring time compared to conventional tactile (contact) measuring instruments, high accessibility to small features on surfaces and low surface-damaging risk.⁶ Optical instruments can collect high-density measuring data from a measured surface part. This data can be leveraged by ML methods to extract more information and reduce measurement time.⁷

The calibration of surface topography measuring instruments is important to achieve metrological traceability to obtain reliable measurement results.⁶ The process can be conducted by following the ISO 25178-600 standard guidelines. The metrological characteristics (MCs) are the calibration properties proposed by the ISO standard for areal surface topography measuring instruments. These MCs are general characteristics that apply to all types of optical measuring instruments (regardless of their operating principle) and directly determine the quality of measurement results.

This paper quantifies the measurement noise of the FVM and applies Bayesian ML via the Gaussian process on the quantified noise data to estimate unknown measurement noise on other locations of the vertical stage of the FVM with different parameters. The measurement noise quantification uses a flat surface artefact to obtain surface data and applies the subtraction method to quantify the noise. It is important to note that different measurement parameters are required to measure different types of surfaces. Measurement noise is significantly affected by the various measurement parameters.

The selected Bayesian ML method via Gaussian process can analyse and predict the result from a small set of data as well as estimate the uncertainty of the predictions. The main contributions of this paper are:

- The quantification of measurement noise with subtraction method for the FVM and extraction of the non-homogenous noise at different height locations of the vertical stage with different measurement parameters.
- The use of the Gaussian process to predict unknown measurement noise at different heights

of the vertical stage of the FVM from a small data set containing known values of measurement noise. This ability can reduce operational costs of measurement (improving overall measurement efficiency toward productive metrology⁸) by reducing the calibration time required to quantify measurement noise.

- The modelling of the Gaussian process and its learning function to optimize the Gaussian process model.
- The optimization of the Gaussian process in which the ML method is used to optimize the hyper-parameter and improve the performance of the measurement noise predictions.
- The selection and arrangement of inputs for the Gaussian process ML model, including application of a hot-encoding method, to increase the prediction accuracy of the ML model.
- The leverage of ML towards a productive metrology paradigm.

The paper is structured as follows: The description of the FVM and the quantification of measurement noise for the FVM measuring instrument is presented in section 2. The estimation of measurement noise for the FVM using Gaussian process ML is covered in section 3. Additionally, the modelling of the Gaussian process ML model is derived in this section. It will also present the Gaussian process ML and hyper-parameter initialization and optimization. The fourth section will include the experiment results and a discussion. Finally, the last section of the paper includes the conclusion and future works.

Focus variation microscopy (FVM) and characteristics of measurement noise

Focus variation microscopy

FVM is areal surface texture measuring instrument that utilizes an optical microscope for areal surface topography measurements.^{6,9} It can perform multiple surface texture measurements for roughness, form, wear and welding spot inspection in both two and three dimensions.¹⁰ These features can be performed as the FVM has a large measurement bandwidth covering high-frequency measurement (roughness) and low-frequency measurement (form). According to the history of the FVM, its early version stage had been launched between 1985 and 1989.^{9,11} Consequently, its final version containing both hardware and software components has been commercially deployed as an instrument for surface topography measurements. This instrument can capture coloured 3D data using specific algorithms to estimate focus from measured surface images.

The working principle of the FVM has some similarities with other optical measuring instruments that obtain the measuring information using the vertical scanning of objective lenses over a measured surface.

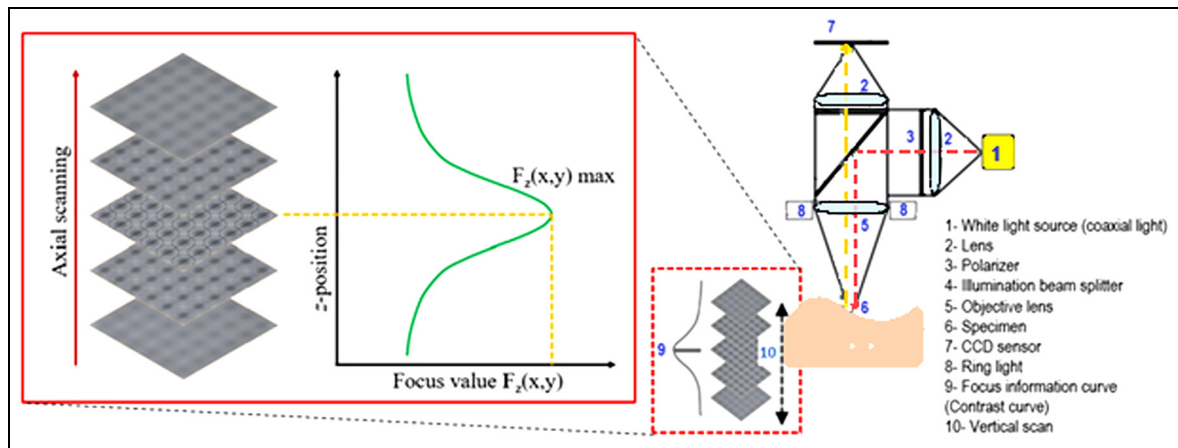


Figure 1. Schematic diagram of FVM and the optical vertical scanning process.¹⁰

The principles are explained in detail elsewhere.^{6,12} Figure 1 shows a schematic diagram of the FVM working principle. In brief, the measurement begins when the emitted white light source passes through an optical tube to reach the splitting mirror. Then, the white light is guided toward the objective lens to measure the surface of the sample. A reflection of light occurs from different directions of a measured surface. The objective lens captures some of the reflected light from the measured surface and passes the light through the optical tube to reach an image sensor. A charge-coupled device (CCD) is a sensor that gathers the reflected light and calculates the value of FV for image forming. Figure 1 indicates a stack of images that can be captured continually along the vertical movement of the optical head to reconstruct the 3D surface. The focus is changed during the movement to affect the contrast on the CCD. The variation of focus for each pixel image is evaluated by a specific algorithm.⁶ Once the position of the focus has been established, the surface height can be measured and the 3D reconstruction data of the measured surface can be obtained.

FVM can detect the surface height at each position depending on the surface image sharpness of the image stack collected during the scan.^{10,13} However, some surface characterization can limit the FVM measurement. The surface texture of 10 nm Ra roughness and high reflection are difficult to measure by typical FVM instruments, as calculating the sharpness of a measured surface is challenging.^{10,14} When the sharpness (focus) is difficult to calculate, the height of the surface cannot be determined. It is found that the replica method could provide an effective process in this measurement situation. The replica is used when measuring extremely smooth and reflective surfaces with the FVM.¹⁵

The FVM in this study is provided by Alicona Imaging GmbH (Bruker–Alicona). The FVM can provide different types of measurement such as dimensions, form, position and roughness. This closes the gap between the 3D coordinates measuring technology and classical tactile surface metrology instruments. The

FVM utilizes a ring light to measure a high slope surface angle that exceeds 80°. The depth information with surface true colour registration can be collected by FVM to enable qualitative and fast surface inspections. Different purposes of activities and applications can be performed using FVM. It can be used in the activity of research, development and industrial quality assurance. Surface measurement using FVM can be used in various applications including the automotive and cutting tool industries, precision manufacturing, development of medical devices, studies related to tribology and corrosion and electronics and material science.¹²

Measurement noise

Measurement noise determination is one of the MCs of areal surface topography measuring instruments. These MCs are recommended by ISO 25178 part 6¹⁶ to calibrate all such instruments and achieve traceability. The measurement uncertainty estimation is part of the calibration process.^{6,17} Besides the measurement noise, the MCs contain flatness deviation, amplification coefficient, linearity deviation, x-y perpendicularity deviation, topographic spatial resolution and topography fidelity. The MC of amplification coefficient, linearity deviation and x-y perpendicularity deviation for FVM has been investigated in Alburayt et al.¹³ Each characteristic has a different determination on the measuring instrument. The aim of applying the MCs on areal surface topography measuring instruments is to provide simple and practical calibration routines that can be applied to different types of optical measuring instruments by the non-expert user.¹⁸ This idea can allow a user in the industry to follow a practical and efficient standard calibration approach. However, one should consider some instrument specifications when applying the MCs of areal surface topography measuring instruments. A reference method of applying the MCs determination is explained in Giusca et al.¹⁹ and Giusca and Leach.²⁰ Each MC can use specific procedures and material measures for the determination. Some

examples include optical flat surface, flat surface and cross-grating artefacts. However, a suitable material measure available for some measuring instruments is still lacking.¹³ Smooth surface artefacts are the most widely used ones in the commercial world. Some measuring instruments such as typical FVM instruments, have some limitations when measuring artefacts with extremely high reflective surfaces. The work of Eifler et al.²¹ has developed a calibrated artefact to determine the MCs. A calibrated artefact was used with different areal surface topography measuring instruments, as a comparison measurement and the uncertainty of early work of comparison results was presented.

The measurement noise is defined in De Bièvre²² as:

Noise added to the output signal occurring during the normal use of the instrument.

The definition clarifies that the noise can be associated with measuring results. The measuring instrument performance and accuracy can be affected by this noise. Thus, the results obtained from the measurement noise can contribute to the uncertainty and traceability of measurement.^{19,23,24} The noise is identified with a high frequency that can limit the measuring instrument's ability to detect small-scale spatial wavelengths of a surface texture.¹² The measurement noise under ideal conditions can be expressed as the instrument noise.²⁵ The ideal conditions mean a constant measuring environment, a perfect surface sample (i.e. reflecting and flat) and ideal sample alignment regarding the measuring axis. Different sources can generate the noise, such as the type of instrument and working area. Measuring in an industrial environment has a high noise level as compared to that from a controlled laboratory.²⁶ The environment surrounding the measuring instrument and the movement of the axis drive during electronic scanning are the sources of external and internal noise, respectively.^{26–28} The environmental sources are temperature and vibration as well as the electronic noise from amplifiers and optical noise. Reducing the output noise can lead to obtaining high accuracy of measuring results. The process of reducing the noise is mentioned in de Groot and DiSciaccia.²⁸ Averaging the field or measuring data over time or smoothing filters that are used can be a method of noise reduction.^{29,30} The measurement noise characteristic can be determined using a measured surface or surface that represents measuring surface types.³¹

The analytical process of the measurement noise can be applied using two techniques: subtraction and averaging methods.^{28,32} These are similar in the output result but different in the calculation method. The user can choose the most efficient analytical method to get the result.²⁷ However, when the calculation results of the two techniques are not similar, the data of a non-stationary surface or the calculation process can be the reason.²⁶ Another analytical process of measurement

noise is applied by using temporal and lateral Allan deviation.³³ This tool is used for deeper noise analysis for surface topography measurements. It can be used for qualification and identification drift and spikes in the mean topography.

The Sq parameter, which is the root mean square of surface height measurements, is used in the estimation of measurement noise (Sq_{noise}).³⁴ The Sq_{noise} is calculated as follows:

$$Sq_{\text{noise}} = \frac{Sq}{\sqrt{2}} \quad (1)$$

The analysis process of obtaining the measurement noise value – is carried out by applying the subtraction method. The (Sq_{noise}) parameter is the average of three (Sq) calculation results to better estimate the noise and reduce the random variations. The measuring data of surface is subtracted and applied to the levelling process – for obtaining (Sq) value. This result of (Sq) is used in the following formula to find the value of (Sq_{noise}). The Sq in the formula is the root mean square after subtraction. The calculation of measurement noise using the subtraction method is performed using Minitab software and MountainsMap surface metrology software for the data analysis procedures.

Gaussian process machine learning for estimating measurement noise of an FVM instrument

Gaussian process machine learning

Gaussian process is a type of Bayesian learning that re-uses training data to perform predictions.³⁵ This learning method is based on the Bayes rule, which is a method to perform inference of a best-fit model to explain observed data.³⁶ By combining the prior knowledge of given data and the likelihood probability of the data, the Bayes rule can estimate the conditional probability of the quantity of a given observed data.³⁷ The quantity is very often the parameter of the hypothesis of a model to explain the data. The Bayes rule is formulated as:

$$P(\theta|D; M) = \frac{P(D|\theta; M)P(\theta; M)}{P(D; M)} \quad (2)$$

Where M is a selected model. D is observed data that uses the model M . $P(\theta|D; M)$ is the posterior probability distribution of the parameters θ after observing data (given the selected model M). $P(D|\theta; M)$ is the likelihood distribution of the data D given the parameters θ in the selected model M . $P(\theta; M)$ is the prior distribution of the parameters (assumed to be auto correlated to each other) that represents previous knowledge about the parameters θ (in the model M) before

observing any data D . $P(D; M)$ is the marginal likelihood (model evidence) of the data D in the selected model M .

Since, $P(D; M) = \int P(D|\theta; M)P(\theta; M)d\theta$, hence equation (1) can be written as:

$$P(\theta|D; M) = \frac{P(D|\theta; M)P(\theta; M)}{\int P(D|\theta; M)P(\theta; M)d\theta} \quad (3)$$

The Bayes rule is used to perform learning to fit the best parameters θ with respect to the selected model M . By, considering given data D as the pair between an input matrix X and an output vector y , the posterior distribution of the θ parameter is proportional to the marginal likelihood as follows:

$$\begin{aligned} P(\theta|D; M) &\approx P(D|\theta; M)P(\theta; M) \leftrightarrow P(\theta|X, y; M) \\ &\approx P(y|X, \theta; M)P(\theta; M) \end{aligned} \quad (4)$$

Equation (3) can be re-written (after some matrix algebraic and completing the square steps) as:

$$\begin{aligned} P(\theta|X, y; M) &\sim N\left(\text{mean } \bar{\theta} = \sigma^2 \left(\sigma^{-2}XX^T + \Sigma^{-1}\right)^{-1} \right. \\ &\quad \left. \times Xy, \text{cov} = \left(\sigma^{-2}XX^T + \Sigma^{-1}\right)^{-1}\right) \end{aligned} \quad (5)$$

Making a new prediction $P(y^*|X^*, X, y; M)$, that is, a Gaussian multivariate distribution, an averaging process over all possible parameters (marginalization process) values θ with respect to a new input $P(y^*|X^*, \theta; M)$ is applied. The parameter values are weighted by their posterior $P(\theta|X, y; M)$. Subsequently, the prediction of $P(y^*|X^*, X, y; M)$ can be calculated as:

$$P(y^*|X^*, X, y; M) = \int P(y^*|X^*, \theta; M)P(\theta|X, y; M)d\theta \quad (6)$$

where X, y are the training data (input-output pairs) that have been observed, X^* and y^* are a new set of inputs and a new prediction, respectively. The prediction of y^* for a given new observed data X^* is:

$$\begin{aligned} P(y^*|X^*, X, y; M) &= N\left(\bar{y}^* = X^{*T}\sigma^2 \left(\sigma^{-2}XX^T + \Sigma^{-1}\right)^{-1} \right. \\ &\quad \left. \times Xy, \text{cov}(\bar{y}^*) = X^{*T} \left(\sigma^{-2}XX^T + \Sigma^{-1}\right)^{-1} X^* \right) \end{aligned} \quad (7)$$

The powerful property of the Gaussian process is that the model complexity can be improved by using a 'kernel trick'. This kernel is used to add the capability of a linear model to be able to model a complex nonlinear pattern from data. With the kernel trick, the

input calculation is performed in high-dimensional space (called kernel space) instead of the input space. Finally, the mean $\bar{y}^*$ and variance of the prediction covariance $\text{cov}(\bar{y}^*)$ become can be calculated as:

$$\bar{y}^* = K(X^*, X)[K(X, X) + \sigma^2 I]^{-1}y \quad (8)$$

$$\begin{aligned} \text{cov}(\bar{y}^*) &= K(X^*, X^*) - K(X^*, X)[K(X, X) + \sigma^2 I]^{-1}K(X, X) \\ &\quad K(X^*, X) \end{aligned} \quad (9)$$

where σ^2 is the input noise, X is a matrix of input used for training, y is a vector of labelled output (with respect to X) used for training and X^* is a matrix of new observed input for prediction.

From the equations (8) and (9), different kernel functions $K(X, X)$ impose different non-linear properties. One important property of kernel $K(X, X)$ is that this kernel should be a positive semi-definite matrix. The kernel represents the similarity in data. Because the basic assumption of the Bayesian (Gaussian) process is that similar data will have similar target values.³⁸ The main advantage of the Bayesian (Gaussian) process is that this model can learn from small datasets and can estimate prediction confidence. This learning capability is very important when learning from a small data set³⁹ as in our case where the number of measurements is not largely available due to multiple experiment constraints, such as expensive samples and limited instrument access.

Hyper-parameter initialization and optimization

The estimation of the best-fit model parameters θ is performed by using an iterative optimization algorithm. In our study, a radial basis function (RBF) or exponential kernel is used. The RBF kernel is formulated as:

$$K[\cdot, \cdot] = \sigma^2 e^{\left(\frac{-\|x_1 - x_2\|^2}{2l^2}\right)} \quad (10)$$

Where σ and l are the kernel parameters to optimize using the iterative optimization algorithm. It is worth noting that since the algorithm is an iterative method with initial solution guessing, the results of the optimization will not guarantee that the optimal parameters will be obtained. The final fit parameters σ and l depend on the optimization parameters, such as the number of iterations, stopping criterion, how good our initial estimates and optimization step λ .

The optimization to find the optimal kernel parameters σ and l is a process to maximize the model evidence, that is, given below:

$$\begin{aligned} \log P(y|X; M) &= -\frac{1}{2}y^T K(X, X)^{-1}y - \frac{1}{2}\log|K(X, X)| - \frac{n}{2}\log 2\pi \end{aligned} \quad (11)$$

Table 1. The results of the measurement noise measurements by using the subtraction method with the various measurement parameters at different vertical heights.

Input							Output
Vertical res. (nm)	Lateral res. (μm)	Image contrast	Exposure time (μs)	Objective lens (×)	Measuring height (mm)	Light type	Sq _{noise} (nm)
10	1.46	1	500	50	20	Coaxial	3.71
10	1.46	1	500	50	40	Coaxial	3.77
10	1.46	1	500	50	60	Coaxial	4.29
10	1.46	1	500	50	80	Coaxial	5.08
10	1.46	1	80	100	20	Coaxial	1.76
10	1.46	1	80	100	40	Coaxial	1.83
10	1.46	1	80	100	60	Coaxial	1.83
10	1.46	1	80	100	80	Coaxial	1.73
10	1.46	1	500	50	20	Polarized	13.86
10	1.46	1	500	50	40	Polarized	14.64
10	1.46	1	500	50	60	Polarized	14.38
10	1.46	1	500	50	80	Polarized	17.23
10	1.46	1	80	100	20	Polarized	3.64
10	1.46	1	80	100	40	Polarized	3.12
10	1.46	1	80	100	60	Polarized	3.48
10	1.46	1	80	100	80	Polarized	4.70
10	1.34	1	142	50	5	Coaxial	7.96
10	1.34	1	142	50	35	Coaxial	6.55
10	1.34	1	142	50	40	Coaxial	7.34
10	1.34	1	142	50	55	Coaxial	7.14
10	1.34	1	142	50	70	Coaxial	7.15
10	1.34	1	142	50	85	Coaxial	7.12
10	1.46	1	142	100	5	Coaxial	.98
10	1.46	1	142	100	35	Coaxial	1.24
10	1.46	1	142	100	40	Coaxial	.99
10	1.46	1	142	100	55	Coaxial	1.13
10	1.46	1	142	100	70	Coaxial	1.11
10	1.46	1	142	100	85	Coaxial	1.19

Where $K[\cdot, \cdot]$ is the kernel matrix that uses the exponential kernel in equation (10) and this matrix has a size that follows the number of elements of the variables of its argument. X is the input variable and y is the output variable (the label). In this case, measurement noise (Sq_{noise}) is presented in Table 1.

For example, $K(x, y)$ has a size of $n \times m$ where n is the number of elements of vector X and m is the number of elements of vector y . Hence, if the two arguments of matrix $K(.,.)$ is the same variables, then $K(.,.)$ is a square $n \times n$ matrix, for example, $K(X, X)$.

The optimization step for σ and l uses a Gauss-Newton step and is formulated as:

$$P^{i+1} = P^i + \lambda \left(\frac{\partial(\log P(y|X; M))}{\partial \theta_i} \right) \quad (12)$$

Where λ is the optimization step in each optimization iteration, θ_i is the i th parameters and

$$\begin{aligned} \log P(y|X; M) \\ = -\frac{1}{2} y^T K(X, X)^{-1} y - \frac{1}{2} \log |K(X, X)| - \frac{n}{2} \log 2\pi \end{aligned} \quad (13)$$

and

$$\begin{aligned} \frac{\partial(\log P(y|X; M))}{\partial \theta_i} &= \frac{1}{2} y^T K(X, X)^{-1} \frac{\partial(K(X, X))}{\partial \theta_i} \\ &K(X, X)^{-1} y - \frac{1}{2} \text{tr} \left(K(X, X)^{-1} \left(\frac{\partial(K(X, X))}{\partial \theta_i} \right) \right) \end{aligned} \quad (14)$$

Where: $\frac{\partial(K(X, X))}{\partial \theta_i}$ is the partial derivation of the exponential kernel.

From equation (11), the higher the number, the search to find better parameter values will be fast, but there is a high probability that the step will by-pass the optimal values, and otherwise.

The partial kernel derivation: $\frac{\partial(K(X, X))}{\partial \theta_i}$ is a kernel K which element is the derivation of the exponential kernel. In this case, the exponential kernel f is defined as:

$$f = \sigma^2 e^{\left(\frac{-\|x_1 - x_2\|^2}{2\sigma^2} \right)} \quad (15)$$

Hence, the partial derivative for each param1 (σ) and param2 (l) is:

$$\frac{\partial f}{\partial l} = \sigma^2 \left[\frac{\|X_I - X_2\|^2}{l^3} \right] e^{\left(-\frac{\|X_I - X_2\|^2}{2l^2} \right)} \quad (16)$$

$$\frac{\partial f}{\partial l} = 2\sigma e^{\left(-\frac{\|X_I - X_2\|^2}{2l^2} \right)} \quad (17)$$

Note that the parameter l is the length of the correlation that describes how long the data is regarded as correlated for making predictions. The longer the value, the smoothing effect on the output prediction will be large, and otherwise.

After optimizing the hyper-parameters, we then can perform predictions by following equations (8) and (9).

To measure the goodness of the fitting or the prediction of the Gaussian process model, mean square error (MSE) and R^2 value metrics are used.

Results of measurement noise and Bayesian learning estimation

Measurement noise quantification for the FVM measuring instrument

A standard Halle flat surface artefact is used in the experiment to quantify the measurement noise of the

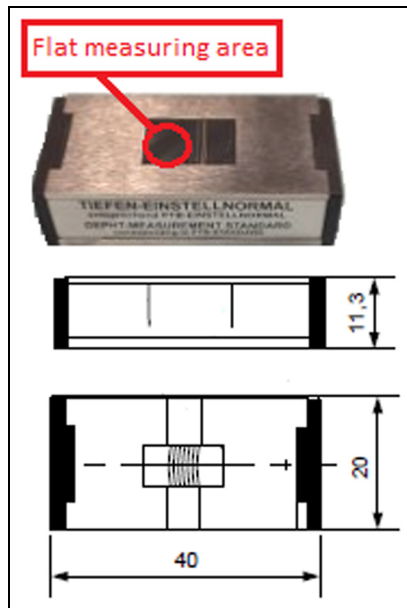


Figure 2. The Halle flat surface artefact. The measuring is conducted on the flat area of the artefact.

FVM. The optical instrument of FVM is used to measure the artefact for determining the measurement noise characteristic. The standard flat surface artefact is stainless steel material with $40 \times 20 \times 11.3$ mm dimension. It has six different depth grooves in the measuring area. However, only the flat areas of the artefact are measured in the experiment. Figure 2 shows the Halle flat surface artefact and its dimension. Figure 2 highlights the measuring area used on the artefact's flat surface.

The experiment uses different measuring parameters. These parameters cover at least 70% of the range of the parameter values that can be set. Seven different measuring parameters are chosen as input in the machine learning. These parameters are vertical resolution, lateral resolution, image contrast, exposure time, objective lens, measuring height and light type. The output data for the machine learning is the (Sq_{noise}) values of the total measurements of different combinations of measurement parameters at different vertical heights.

The details of measuring parameters are in Table 2. The reasons for the parameters and their value selections are as follows:

- The parameters are the fundamental measurement parameters for focus variation microscopy.⁶
- The parameters cover the range of the values $> 70\%$.
- The parameters represent the typical range of surface topography measurement using FVM.¹³
- The objective lenses are selected to be able to faithfully reconstruct high-frequency details features on surface topography instead of the low-bandwidth features (form) of the surfaces.⁶

The data of vertical resolution and image contrast parameters for all measurements are constant. Changing the value of imaging contrast can lead to different measuring times. Making the value constant can assist in obtaining good measuring data in a short time. The data of vertical resolution and image contrast parameters are 10 nm and 1, respectively.

Two different lateral resolutions of 1.46 and $1.34 \mu\text{m}$ are used in the experiment. Three different inputs of exposure time have been determined in the measurement. Different settings of exposure time are applied to make the diversity in the image bright. The exposure time in the measurement is 80, 142 and $500 \mu\text{s}$. The

Table 2. The measurement parameters (factors) and their values (level).

Measurement parameter (factor)	Value (level)	Description
Vertical resolution (nm)	10	Quantitative
Lateral resolution (μm)	1.34, 1.46	Quantitative
Image contrast	1	Quantitative
Exposure time (μs)	80, 142, 500	Quantitative
Objective lens ($\times$)	50, 100	Quantitative
Measuring height (the position of the vertical stage; mm)	5, 20, 35, 40, 55, 60, 75, 80, 85	Quantitative
Light type (illumination)	Coaxial, polarized	Qualitative

measurement data has been collected in the experiment by two objective lenses of $50\times$ and $100\times$, respectively. The vertical heights of the measuring instrument are moving vertically to measure the artefact. The vertical height moves to 100 mm. The vertical heights used in the experiment are 5, 20, 35, 40, 55, 60, 70, 80 and 85 mm. Figure 3 shows the vertical heights of FVM. The two different settings of the light types are normal coaxial light and polarizer light. Twenty repeated



Figure 3. The vertical height of FVM. The red circle presents the vertical height.

measurements are carried out for each measuring setting. This repetition of measurement can assist in applying averaging and reducing the random error effects.

The analysis process for obtaining the measurement noise value has been performed in this experiment by applying the subtraction method.^{26,31} Two different measurement data of the same measured surfaces are subtracted from each other. This analytical subtraction method uses the MountainsMap software program. This method removes the systematic elements from the two surfaces and then extracts the random element, the noise, from the residual of the subtracted two surfaces. Then, the levelling process is applied to the subtracted measuring surface to remove the element of linear form from the subtracted two surfaces. Following the subtraction process, a threshold process is applied on the levelled surface to remove outliers in data. Outliers are common in measured data, that is, obtained from optical instruments since they are commonly caused by specular reflections from a measured surface or electronic noises in the instrument.^{5,6}

After the removal of outliers, the Sq parameter is calculated. This Sq is calculated as the root mean square of the measured surface data after the subtraction, levelling and outlier removal processes. Finally, from the calculated Sq parameter, the measurement noise of the measured surface is calculated following equation (1).

Figure 4 presents the step-by-step analysis process to quantify the measurement noise of measured flat surfaces from the artefact using the MountainsMap

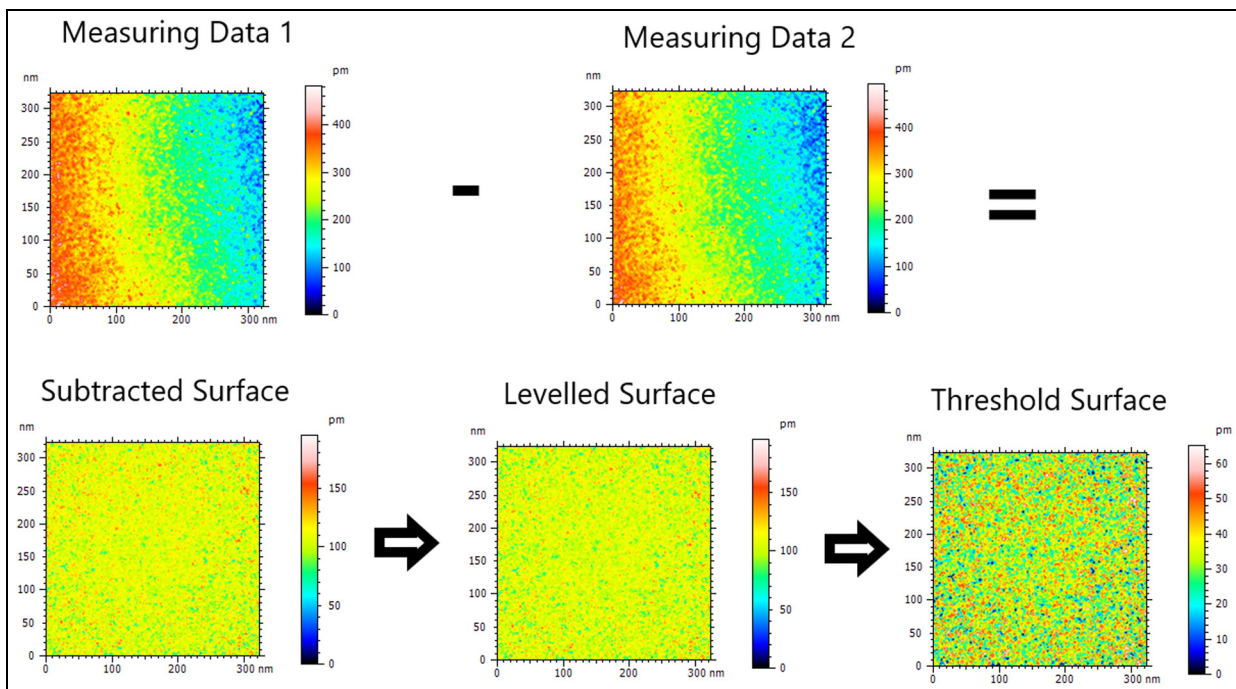


Figure 4. The step-by-step process of the subtraction method to estimate the measurement noise of focus variation microscopy with the flat artefact.

Table 3. The final input and output variables for the ML model and their data type. After implementing the hot-encoding method, the light type is separated into polarized and coaxial fields.

Input (X)								Output (y)
Vertical res. (nm)	Lateral res. (μm)	Image contrast	Exposure time (μs)	Objective lens ($\times$)	Measuring height (mm)	Polarized	Coaxial	Sq_{noise} (nm)
Float	Float	Float	Float	Float	Float	Integer	Integer	Float

software program. From Figure 4, two measurement data from the identical area on the flat surface have been subtracted from each other. Following this subtraction, the levelling process is applied. From this levelled surface, the thresholding method for outlier removal is used to calculate the Sq parameter and Sq_{noise} .

The obtained Sq_{noise} value, the measurement noise estimation, from the subtraction method is calculated for three times on the same surfaces. These three calculations of Sq_{noise} value are averaged to better estimate the measurement noise and reduce variation on the estimated measurement noise.

The results of the quantification of measurement noise with the various measurement parameters at different heights of the vertical stage are presented in Table 1. From Table 1, since each measurement noise is calculated from the average of three noise estimations and each measurement noise calculation requires two consecutive surface measurements, a total of 168 surface measurements of the flat artefact have been carried out. From these total surface measurements, a total of twenty-eight measurement noise data values, as being presented in Table 1, are calculated at different vertical stage location and measurement parameters.

Measurement noise estimation by using Bayesian learning

The Bayesian machine learning method is implemented on the available data as shown in Table 2. This data is regarded as the training data and represents $>60\%$ of the vertical range of the FVM instrument. This data is deemed enough considering the ability of the Bayesian method to learn from a relatively limited set of data as well as to give uncertainty on predictions. In addition, a relatively small set of data will improve the prediction computational efficiency of the Bayesian method. The Bayesian learning will re-use all training data when performing predictions as described in equations (8) and (9).

The input and output of the ML model are shown in Table 3. Here, it can be observed that ‘light type’ (in Table 2) is qualitative data. Hence, the ‘hot-encoding’ method is applied to this input so that the input becomes two separate fields ‘Polarized’ and ‘Coaxial’ with values of either 0 or 1.

From all the data sets shown in Table 1, the data are separated into training and testing datasets. Table 4

shows the division between the training and testing data. It is important that the testing data are not included in the training data. In Table 4, the training data is used to optimize (learn) the kernel hyper-parameter (equation (10)) by maximizing the model evidence (equation (11)) so that the model can make predictions accurately (compared to the testing data in Table 4).

The learning process is performed by optimizing, in this case maximizing, the hyper-parameter of the kernel. The hyper-parameter solution space is determined by calculating the model evidence for possible values of the hyper-parameter as shown in equation (11). Figure 5 shows the plot of the hyper-parameter solution space at different values of the hyper-parameter. From this figure, it can be observed that the solution space is continuous.^{36,39} However, the hyper-parameter solution space being non-linear, an iterative optimization approach based on Gauss–Newton method is used to find the optimal value of the hyper-parameter of the Gaussian process ML model.

For the optimization process, since it is iterative, an initial solution is required. The initial solutions for the two parameters of the hyper-kernel are given as (param 1) $\sigma = 1.5$ and (param 2) $l = 1.5$. The stopping criteria of the optimization process are that if there is no improvement on the model evidence from the previous steps or if the iterations have reached the maximum number specified which is set to 100. These initial values will be improved along the optimization steps.

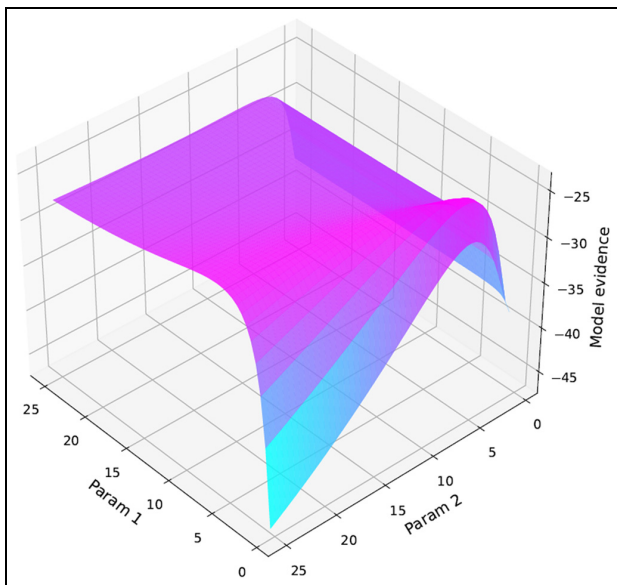
By implementing the Gauss–Newton optimization process (equations (12)–(14)) to learn the kernel’s hyper-parameters, the learned hyper parameters are (param 1) $\sigma = 2.744$ and (param 2) $l = 10.1$. The training accuracy is 96.4% with an average training uncertainty of 0.08 nm.

For predictions with the Bayesian estimation, these can reach an accuracy of 94% with limited training data with an average uncertainty of 0.076 nm. The predictions use all the inputs provided in Table 3. Table 5 shows the prediction results including their uncertainty. Two parameters are included in this table to clearly explain the data.

Figure 6 shows the comparison plot between the target and predicted noise value. As can be observed in this figure, the measurement noise is significantly reduced for measurement using a $100\times$ objective lens. All target noise values are within the 95% confidence

Table 4. The grouping of the training and testing data for the Bayesian estimation of measurement noise.

Training data (re-used for future predictions or testing)							Output (y)
Input (X)							$S_{q_{noise}}$ (nm)
Vertical res. (nm)	Lateral res. (μm)	Image contrast	Exposure time (μs)	Objective lens ($\times$)	Measuring height (mm)	Light type	
10	1.46	I	500	50	20	Coaxial	3.71
10	1.46	I	500	50	40	Coaxial	3.77
10	1.46	I	500	50	60	Coaxial	4.29
10	1.46	I	500	50	80	Coaxial	5.08
10	1.46	I	80	100	20	Coaxial	1.76
10	1.46	I	80	100	40	Coaxial	1.83
10	1.46	I	80	100	60	Coaxial	1.83
10	1.46	I	80	100	80	Coaxial	1.73
10	1.46	I	500	50	20	Polarized	13.86
10	1.46	I	500	50	40	Polarized	14.64
10	1.46	I	500	50	60	Polarized	14.38
10	1.46	I	500	50	80	Polarized	17.23
10	1.46	I	80	100	20	Polarized	3.64
10	1.46	I	80	100	40	Polarized	3.12
10	1.46	I	80	100	60	Polarized	3.48
10	1.46	I	80	100	80	Polarized	4.70
10	1.34	I	142	50	40	Coaxial	7.34
10	1.34	I	142	50	70	Coaxial	7.15
10	1.46	I	142	100	40	Coaxial	.99
10	1.46	I	142	100	70	Coaxial	1.11
Testing data							
Input (X*)							Output (y*)
Vertical res. (nm)	Lateral res. (μm)	Image contrast	Exposure time (μs)	Objective lens ($\times$)	Measuring height (mm)	Light type	$S_{q_{noise}}$ (nm)
10	1.34	I	142	50	5	Coaxial	7.96
10	1.34	I	142	50	35	Coaxial	6.55
10	1.34	I	142	50	55	Coaxial	7.14
10	1.34	I	142	50	85	Coaxial	7.12
10	1.46	I	142	100	5	Coaxial	.98
10	1.46	I	142	100	35	Coaxial	1.24
10	1.46	I	142	100	55	Coaxial	1.13
10	1.46	I	142	100	85	Coaxial	1.19

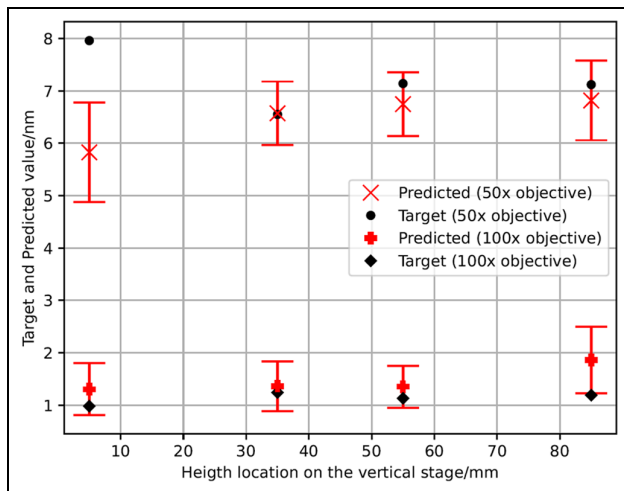
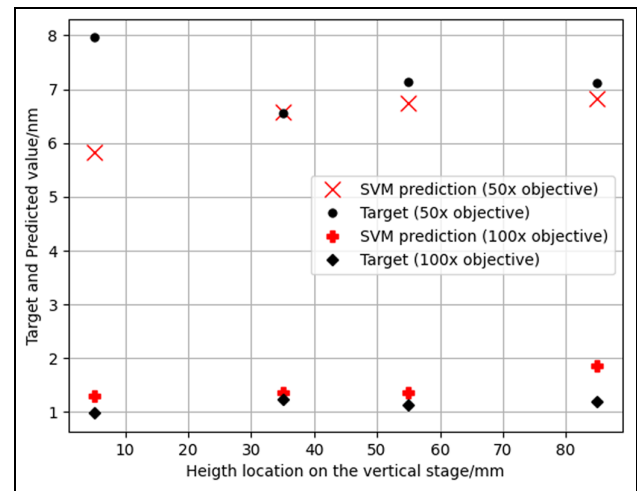
**Figure 5.** The hyper-parameter optimization search space.

interval of the prediction, except for the target noise value at 5 mm vertical height and the 50 $\times$ objective lens.

Specifically for the noise prediction at 5 mm vertical height and the 50 $\times$ objective lens, the prediction of the measurement noise is outside the prediction uncertainty limit. This inaccurate prediction for this value at 5 mm height means that the variation property of the noise in this specific position and objective lens is quite different as compared to other noise. Additionally, this noise is at the boundary of the model.^{36,38} This particularly large noise could be related to low position repeatability (high position variation) when a vertical scanning over the surface is performed. This low repeatability may be caused by mechanical or component errors on the vertical stage at this location, that is, close the vertical stage limit.⁴⁰ Particularly, bearing and shaft tensions close to its end of a motion stage is higher than that far from the limit or end of the stage.⁴⁰ From these results, it is recommended that part measurement should be performed at a height

Table 5. The prediction and target measurement noise with some measurement parameters are shown in the table.

Objective lens	Measuring height (mm)	Target noise value (nm)	Predicted noise value (nm)	Uncertainty (nm)
50×	5	7.96	5.83	0.47
50×	35	6.55	6.57	0.30
50×	55	7.14	6.75	0.30
50×	85	7.12	6.82	0.38
100×	5	0.98	1.30	0.25
100×	35	1.24	1.36	0.24
100×	55	1.13	1.35	0.20
100×	85	1.19	1.86	0.32

**Figure 6.** The predictions of the measurement noise with the Bayesian estimation method with their prediction uncertainty.**Figure 7.** The predictions of the measurement noise with SVM regression method with linear kernel.

location, that is, not close to or near the end limit of the vertical stage of the instrument.

For further comparison, we compare the noise prediction with support vector machine (SVM) regression model. The comparison parameter is the root mean-squared error (RMSE) of the predictions. With the Bayesian method, the RMSE of the prediction is 0.3 nm. Meanwhile, the RMSE of the SVM regression is 1.8 nm. Furthermore, the SVM prediction cannot provide prediction uncertainty as those provided by the Bayesian method. Figure 7 shows the prediction of the SVM regression. From this figure, since we do not have prediction uncertainties, it will be not reliable to compare the prediction and the target values. From these results, the Bayesian method show advantages in term of prediction accuracy and reliability (by providing the prediction uncertainty).

It is important to note that measurement noise is one metrological characteristic and is one of uncertainty contributor for the total uncertainty of a surface topography measurement. Hence, in practice only the average value of a measurement noise is considered to be added as one of the surface topography measurement uncertainties.³¹ The uncertainty in the prediction of the model (shown in Figure 6) is the standard

deviation from the Bayesian model calculated from the variation within the dataset and is used to make sure that model prediction is within a certain range. However, the mean value of the prediction is the one used as one of the uncertainty contributors (one of metrological characteristics) of the total surface topography measurement uncertainties.

Conclusion and future works

Measurement noise is one of the main uncertainty contributors to the results of optical surface topography measurement. In this paper, the quantification of measurement noise and a method to predict the measurement noise of FVM have been presented and discussed. The proposed prediction method uses the Gaussian process ML with an exponential kernel. The main goal is to predict unknown measurement noise in other locations on the vertical stage of an FVM measurement, given the measurement parameters. This objective can be achieved by considering certain measurement parameters like vertical resolution, lateral resolution, image contrast, exposure time, objective lens, measurement vertical position with respect to the vertical motion stage axis and type of illuminations (light sources).

The Gaussian process ML model can give sufficiently accurate measurement noise estimates at different height locations across the vertical stage of the FVM and with different measurement parameters. Only a small set of known measurement noise is required. This ML prediction can significantly reduce calibration time thus reducing the total measurement cost. Without this ML prediction, measurement noise needs to be quantified for all possible measurement height locations and different measurement parameters, as different surface types and part shapes will require different height locations and measurement parameters.

The process of parameter optimization of the exponential kernel of the Gaussian process model has been presented and discussed. This optimization is based on the gradient descent method. The objective function, which is to be maximized to get the best parameter fit, is presented by maximizing the model evidence. The derivation of the parameters for the optimization step has also been presented.

From the results, the measurement noise can be predicted, given the measurement process parameters using the mean squared error of 0.064 and with a goodness of fit $R^2 = 0.96$. With this prediction, the best measurement parameters can be selected beforehand to minimize the measurement uncertainty of the FVM measurement results.

Future work will be to extend the ML model to predict other measurement uncertainty contributions for optical measurement of surface topography, such as flatness deviation and applying the measurement noise prediction for other optical instruments, such as coherence scanning interferometry and confocal microscopy.

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Data availability

The data of this work will be available upon request from the corresponding author.

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